IDSC 4101: Generative and Agentic AI for Business

Course Syllabus

Course Information

Course Title: Generative and Agentic AI for Business
Course Number: IDSC 4101
Credits: 2 credit hours
Meeting Pattern: Two sessions per week, 100 minutes per session
Duration: 7 weeks (14 total sessions)
Prerequisites: Junior or Senior standing; any business major or concentration
Coding Required: None

Instructor Information

Instructor: [Instructor Name]
Office: [Building and Room Number]
Office Hours: [Days and Times] — walk-in and by appointment
Email: [Institutional Email]
Response Policy: Instructor will respond to emails within 48 hours on business days. For urgent questions, office hours are preferred.

Teaching Assistant: [TA Name, if applicable]
TA Office Hours: [Days and Times]

Course Description

Artificial intelligence is no longer a technology story — it is a business transformation story. This course equips upper-division business students with the conceptual foundations, strategic frameworks, and hands-on fluency to understand, evaluate, and deploy generative and agentic AI in organizational contexts. Moving well beyond prompt engineering, students will develop working knowledge of how large language models function, how AI agents perceive and act on their environments, and how no-code automation platforms enable businesses to redesign workflows, products, and services without writing a single line of code.

Over seven weeks, students progress from foundational AI literacy through applied automation and agentic system design, culminating in a team capstone that demonstrates a real business use case. The course integrates tools from the Google Gemini ecosystem, no-code builders such as Zapier, Make, and Relevance AI, and agentic platforms including n8n and Voiceflow. Students will also engage with the strategic and ethical dimensions of AI adoption — workforce impact, governance, vendor selection, and responsible deployment — preparing them for roles in which AI fluency is increasingly a baseline expectation regardless of functional area.

This course is designed for students across all business concentrations — marketing, finance, management, operations, entrepreneurship, and supply chain — and requires no prior technical or programming background.

Course Rationale

The business landscape is undergoing a structural shift. Generative AI tools — large language models, image generators, and multimodal systems — have moved from novelty to operational infrastructure in under three years. Agentic AI, in which AI systems plan, take actions, and coordinate with other systems autonomously, is accelerating this shift further. According to McKinsey's 2025 State of AI report, over 70 percent of organizations are piloting or deploying generative AI in at least one business function. Roles in marketing, finance, HR, and operations now routinely require AI fluency as a baseline skill.

Yet most business graduates enter the workforce with only surface-level exposure — typically limited to using a chatbot interface. They lack the conceptual vocabulary to evaluate AI vendor claims, the workflow intuition to identify automation opportunities, and the strategic grounding to lead or contribute to AI initiatives. This course closes that gap. It is explicitly designed for business majors who will manage AI tools, commission AI projects, and make strategic decisions about AI adoption — not for those who will build AI systems from scratch. The goal is informed, confident fluency: the ability to understand what AI can and cannot do, translate business problems into AI solutions, and lead responsibly in AI-augmented organizations.

Learning Objectives

By the end of this course, students will be able to:

1. **Explain** the foundational mechanics of large language models and generative AI systems — including tokenization, training, inference, and hallucination — using accurate, non-technical language appropriate for business communication.
2. **Distinguish** among AI paradigms (rule-based systems, predictive ML, generative AI, and agentic AI), identifying which paradigm is appropriate for a given business problem and what constraints each carries.
3. **Design and execute** no-code AI-assisted workflows using automation platforms, connecting AI outputs to real business processes in marketing, operations, finance, or HR without writing code.
4. **Evaluate** agentic AI systems — including multi-agent architectures, tool-using agents, and AI copilots — by analyzing their reasoning patterns, failure modes, and appropriate scope of autonomy within an organizational context.
5. **Assess** the strategic, ethical, and governance implications of AI adoption in a business context, including workforce impact, bias and fairness risks, regulatory considerations, and vendor evaluation frameworks.
6. **Apply** a structured business problem-solving process to identify, scope, prototype, and present an AI-assisted solution to a real organizational challenge using tools covered in the course.

Prerequisites

- Junior or Senior standing required

- Any business concentration accepted (marketing, finance, management, operations, supply chain, entrepreneurship, or undeclared)
- No prior coursework in computer science, data science, or programming required or assumed
- Basic comfort with cloud-based software (Google Workspace, Microsoft 365, or equivalent) is helpful but not required
- Students are expected to bring intellectual curiosity and willingness to experiment with emerging tools

Recommended (not required): One prior course in business strategy, organizational behavior, or management information systems will provide helpful context for weeks 5–7.

Required Tools and Platform Access

This course uses no-cost or student-tier access to the following categories of tools. No software purchase is required; all platforms have free tiers sufficient for coursework. Specific setup instructions and access links will be distributed in the first week.

Conversational AI and Multimodal AI (Weeks 1–4)
The Google Gemini ecosystem — including Gemini (conversational AI), NotebookLM (AI-powered research assistant), and Google AI Studio (prompt and model experimentation) — serves as the primary AI interface for the first half of the course. Students will use these tools to understand how large language models respond to different types of prompts, how context shapes output, and how multimodal inputs (text, image, document) change AI behavior. All tools are accessible via a standard Google account. Students without a Google account will need to create one before the first session.

No-Code AI Workflow Builders (Weeks 3–5)
Platforms in this category — including Zapier, Make (formerly Integromat), and Relevance AI — allow users to connect AI capabilities to other software systems (email, spreadsheets, CRMs, calendars, databases) through a visual drag-and-drop interface. Students will use these tools to build automated workflows that incorporate AI decision-making without writing code. Free tiers on all three platforms are sufficient for course assignments. Students will choose one platform to use for their capstone project.

Agentic AI and Conversational Agent Platforms (Weeks 5–6)
Platforms in this category — including n8n (workflow automation with agentic logic) and Voiceflow (conversational agent builder) — enable the construction of multi-step, goal-directed AI workflows and dialogue-based agents. These tools represent the frontier of accessible, no-code agentic AI. Students will use these platforms to build and test simple AI agents that can take sequential actions, branch based on outputs, and interact with users. Free tiers or educational access will be confirmed at course start.

Note on Tool Evolution: The AI tool landscape evolves rapidly. The platforms listed above are current as of course launch; the instructor reserves the right to substitute equivalent tools if access, pricing, or functionality changes materially. The conceptual skills developed in this course — workflow design, agent architecture, prompt strategy, governance evaluation — transfer across specific platforms.

Required Materials

This course does not use a textbook. The AI field moves faster than print publishing cycles allow. Instead, required readings draw from three sources:

1. **Assigned articles and reports** — A curated reading list of practitioner reports (McKinsey, Deloitte, NIST), academic summaries, and long-form journalism will be distributed via the course LMS each week. Readings average 30–45 minutes per session.
2. **Case studies** — Harvard Business School and MIT Sloan case studies on AI adoption, automation strategy, and AI ethics will be used in weeks 3, 5, and 7. Case PDFs will be distributed through the course LMS. If your institution charges for HBS cases, a small fee (typically $3–5 per case) may apply; the instructor will confirm before week 3.
3. **AI-generated and tool-generated readings** — In some weeks, students will use NotebookLM or Gemini to generate synthesized briefings on topics, then critically evaluate those outputs as part of the learning activity. This is intentional: reading AI-generated content critically is itself a core course skill.

Recommended (Optional):
- *Co-Intelligence: Living and Working with AI* — Ethan Mollick (2024): The most accessible book-length treatment of working with AI for non-technical professionals.
- *The Coming Wave* — Mustafa Suleyman (2023): Strategic and geopolitical context for AI's trajectory.

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7-Week Session Schedule

How to Read This Schedule

Each week has a unifying theme and two 100-minute sessions. **Session 1** typically introduces concepts and tools through demonstration and guided exploration. **Session 2** builds on that foundation with a hands-on activity, case discussion, or team build sprint. Readings listed under each week should be completed **before** Session 1 unless otherwise noted. Tools are introduced progressively — do not worry about mastering earlier tools before new ones appear; the course is designed to layer skills.

The arc of the course follows a deliberate progression:

- **Weeks 1–2:** What is generative AI, and how does it already show up in your life? (Personal productivity lens)
- **Weeks 3–4:** How do businesses use this? What can I build for a team or organization? (Workflow and business application lens)
- **Weeks 5–6:** What are AI agents, and how do businesses deploy them as workers, assistants, and decision-makers? (Agentic and automation lens)
- **Week 7:** Where is this going, and what does it mean for how I work, lead, and make decisions? (Strategy, ethics, and responsibility lens)

Week 1: Welcome to the AI Era — What Is Generative AI and Why Does It Matter?

Theme: Orientation, curiosity, and first contact with generative AI tools.

Session 1 — The New Landscape (Day 1)

Topics:
- Course orientation: what we will build, what we will understand, how we will be assessed
- What generative AI actually is — and is not: distinguishing it from search engines, calculators, and traditional software
- A brief, non-technical mental model: how AI learns from text and generates responses (no math; think of it as a very well-read assistant that predicts what to say next)
- The generative AI moment: why 2022–2025 was a hinge point in technology history
- Live demo: instructor uses Gemini live in class to answer a business question, draft a memo, and make a factual error — students observe and discuss

Activity: "First Contact" — students open Gemini for the first time and complete three tasks: ask a question in their area of interest, give the AI a bad prompt, then a better one, and compare outputs. No setup beyond a Google account.

Discussion: What surprised you? What did it get right? What did it get wrong?

Tools Introduced: Google Gemini (conversational interface)

Session 2 — Personal AI: Making Your Own Work Life Better

Topics:
- Generative AI as a personal productivity tool — not a replacement for thinking, but an accelerant
- Use cases every business student can use today: drafting emails, summarizing documents, preparing for interviews, building study guides, researching companies before a job application
- Prompting as communication: the difference between a bad prompt and a great one is not magic — it is clarity, context, and specificity
- Introduction to prompt patterns: role prompting, step-by-step instructions, asking for multiple options, giving examples

Activity: "Upgrade Your Workflow" — each student picks one recurring task from their academic or personal life (weekly recap email, meeting agenda, study notes, cover letter draft) and uses Gemini to build a reusable prompt template for it. Students share their best templates with the class.

Readings (before Session 1):
- Mollick, E. "Using AI to Make Work More Meaningful" — *Harvard Business Review*, 2024
- Assigned briefing: "What is a Large Language Model?" (NotebookLM-generated summary, provided via LMS — students will evaluate it critically in class)

Assignment Due: None (week 1 is orientation)

Week 2: Going Deeper — Understanding AI Well Enough to Use It Confidently

Theme: Building just enough conceptual foundation to be an informed, critical user — not a builder of models, but someone who understands why AI behaves the way it does.

Session 1 — Why Does AI Do That?

Topics:
- Tokens, context windows, and why AI "forgets" things in long conversations
- Hallucination: what it is, why it happens structurally, and how to reduce it — this is not a bug to be fixed, it is a feature of how the technology works
- Temperature and creativity: why AI gives different answers to the same question
- Multimodal AI: what changes when AI can see images, read PDFs, and hear audio
- Model differences: what it means when a tool says it uses a "more capable" or "faster" model, and how to think about choosing tools

Activity: "Break the AI" — structured red-teaming exercise. Teams are given a list of prompting strategies (adding false context, asking the AI to roleplay, pushing it toward confident wrong answers) and test Gemini's behavior. Teams report findings. Discussion: what does this tell us about how to use AI responsibly?

*Session 2 — NotebookLM: AI as a Research and Knowledge Partner**

Topics:
- Introduction to NotebookLM: AI that reads your documents and lets you ask questions about them
- Use cases: summarizing a 40-page report, extracting key arguments from a case study, generating a briefing document for a client meeting, building a study guide from lecture notes
- Limits: NotebookLM only knows what you give it — and it can still be wrong about what it reads
- Ideation use case: using AI to pressure-test a business idea by uploading competitor websites, industry reports, and your own notes

Activity: "The Analyst Sprint" — each student uploads a set of documents relevant to their concentration (a 10-K, a case study, an industry report — provided) to NotebookLM and builds a one-page executive briefing by querying the AI. Teams compare their briefings and identify discrepancies.

Tools Introduced: Google NotebookLM, Google AI Studio (first look — used for prompt experimentation)

Readings (before Session 1):
- Assigned article: "The Hallucination Problem and What to Do About It" — *MIT Technology Review*, 2024
- Optional: Suleyman, M. *The Coming Wave*, Chapter 1 (recommended text)

Assignment Due: **Reflection Post 1** — 200–300 words: Describe one moment this week when the AI surprised you (positively or negatively). What does that surprise tell you about how the tool works?

Week 3: AI for Teams — Automating the Work Nobody Wants to Do

Theme: From personal tools to team workflows. Students learn to connect AI to the systems businesses already use — without writing code.

Session 1 — Workflow Thinking: What Gets Automated and What Does Not

Topics:
- The automation opportunity: which business tasks are repetitive, rule-based, and time-consuming enough to automate?
- A framework for identifying automation candidates: frequency × effort × standardizability
- Introduction to no-code workflow builders: what they are, how they work visually, what they can connect (email, spreadsheets, CRMs, Slack, databases, AI models)
- Demonstration: building a simple Zapier workflow that takes a form submission, passes it to Gemini for a summary, and emails the result — live in class, no code

Activity: "Automation Audit" — in teams, students map the workflows of a hypothetical small business (provided scenario) and identify the top 3 candidates for AI-assisted automation. Teams present their reasoning using the frequency × effort × standardizability framework.

*Session 2 — Build Sprint: Your First Automation**

Topics:
- Getting set up with Zapier or Make (students choose their preferred platform)
- The anatomy of a no-code workflow: triggers, actions, filters, and AI steps
- Common business automations: lead qualification, meeting follow-up drafting, customer feedback summarization, content repurposing, invoice processing
- Relevance AI: building AI-powered assistants that follow business-specific rules

Activity: **Build Sprint 1** — each student builds a working automation connecting at least two tools with an AI step in the middle. Examples: a Google Form that summarizes responses with Gemini and posts to a shared Slack channel; an email inbox watcher that drafts replies using AI. Instructor circulates; peer help encouraged.

Tools Introduced: Zapier (or Make), Relevance AI (introduction)

Readings (before Session 1):
- McKinsey & Company: "The State of AI in 2025" — assigned excerpt (provided via LMS)
- Case study: "Automation at Scale: How [Company] Redesigned Its Customer Service Workflow" (HBS case — to be confirmed)

Assignment Due: **Reflection Post 2** — 200–300 words: What workflow did you build? What broke, and how did you fix it? What would you automate next?

Week 4: Business Applications by Function — AI Across the Organization

Theme: Industry and functional applications. Students explore how AI is being used in their own concentration areas and develop an ideation vocabulary for new applications.

Session 1 — AI in Marketing, Sales, and Customer Experience

Topics:
- AI in marketing: content generation, audience segmentation, personalization at scale, A/B testing with AI assistance
- AI in sales: lead scoring, outreach personalization, call summarization, pipeline forecasting
- AI in customer experience: chatbots vs. AI agents, sentiment analysis, support ticket routing
- The "ideation mindset": how to look at a business function and ask "where is the AI opportunity?" — not "how do I build it?" but "what would it need to do?"
- Guest perspective (video or live): practitioner in marketing or sales describing their AI toolkit

Activity: "Function Mapping" — each student maps their intended career function and identifies 3 concrete AI use cases. They pitch the highest-impact one to a partner, who challenges it with a "what could go wrong?" critique.

Session 2 — AI in Operations, Finance, and HR

Topics:
- AI in operations: demand forecasting, logistics optimization, quality control flagging, supplier communication
- AI in finance: financial report summarization, anomaly detection, automated client briefings, regulatory

document review
- AI in HR: job description writing, resume screening augmentation, onboarding content generation, employee sentiment monitoring
- Cross-functional theme: AI as a force multiplier — small teams doing more, not headcount replacement (nuanced discussion)

Activity: **Midterm Exam Review Session** — instructor-led review of core concepts from weeks 1–4 using live class polling. Students submit 2 questions they are unsure about; instructor addresses top questions. Remainder of session: team capstone brainstorm — teams of 3–4 form and begin identifying their capstone business problem.

Readings (before Session 1):
- Assigned report: "AI Adoption by Business Function" — Deloitte AI Institute excerpt (provided via LMS)
- Assigned article: "AI Won't Replace Humans — But Humans with AI Will Replace Humans Without AI" — *HBR*, 2023

Assignment Due: **Midterm Conceptual Exam (take-home, open-note, individual)** — released after Session 2, due before Week 5 Session 1. Format: 4 case-based short-answer questions testing conceptual understanding of LLMs, workflow automation, and functional AI applications. No coding. Approximately 90 minutes.

Week 5: Agentic AI — Building Systems That Act, Not Just Answer

Theme: Crossing from generative AI (AI that responds) to agentic AI (AI that acts). Students learn what agents are, how they reason, and how businesses are beginning to deploy them.

Session 1 — What Is an AI Agent?

Topics:
- The key difference: generative AI responds; agentic AI plans, takes actions, and loops back
- Agent anatomy without code: goals, tools, memory, reasoning loops — explained through analogy and diagram
- Types of agents students will encounter in business: task agents, research agents, customer-facing conversational agents, multi-agent systems (one agent manages others)
- Agentic AI design patterns: reflection (agent checks its own output), tool use (agent calls external systems), planning (agent breaks a goal into steps), handoff (agent routes to human or another agent)
- Real business examples: AI SDR (sales development representative) that researches, drafts, and sends outreach; AI operations agent that monitors inventory and flags exceptions; AI onboarding agent that answers new employee questions

Activity: "Agent Blueprint" — teams sketch an agent for a business use case they find interesting. They define: What is the agent's goal? What tools does it need access to? When should it hand off to a human? What could go wrong?

Session 2 — Build Sprint: Your First Agent with n8n or Voiceflow

Topics:
- Introduction to n8n and Voiceflow: what each is best for (n8n for workflow-based agents; Voiceflow for conversational agents)
- Building a simple agent: a customer FAQ bot that can look up answers, escalate when it does not know, and log interactions
- Introduction to the concept of "guardrails": how to scope what an agent is and is not allowed to do

Activity: **Build Sprint 2** — teams build a simple agent using n8n or Voiceflow. Suggested builds: a customer service agent for a fictional business, a meeting scheduler agent, or a research agent that searches and summarizes. Teams demo their agent to another team at the end of the session.

Tools Introduced: n8n, Voiceflow

Readings (before Session 1):
- Assigned article: "The Rise of AI Agents in Enterprise" — *MIT Technology Review*, 2025 (provided via LMS)

Assignment Due: **Reflection Post 3** — 200–300 words: What does your agent do? What decision did you make about what it should NOT be allowed to do, and why?

Week 6: Building for Real — AI in Products, Ventures, and Business Models

Theme: From individual tools to business-level thinking. How do companies build AI into their products and business models? How do you think like an AI product designer without being an engineer?

Session 1 — AI-Native Products and Business Models

Topics:
- What makes a product "AI-native" vs. "AI-enhanced"?
- Business model patterns enabled by AI: mass personalization, AI-as-a-service, outcome-based pricing, data flywheels
- Startup and enterprise examples: companies that are fundamentally different because of generative AI at their core
- The ideation toolkit: from user problem → AI capability → workflow design → value proposition — walking through the product thinking loop without needing to write code
- Vendor evaluation: how do you choose an AI platform? Key questions: data privacy, cost at scale, customizability, reliability, vendor lock-in risk

Activity: "AI Product Pitch" — teams pitch a hypothetical AI-native product or service to the class (2 minutes each). Peers vote on feasibility and impact. Instructor gives feedback on product thinking.

Session 2 — Responsible AI: Ethics, Security, and What Can Go Wrong

Topics:
- Why responsible AI matters in practice, not just in principle: reputational, legal, and operational risks
- Bias and fairness: where bias enters AI systems and how to spot warning signs without being a data scientist
- Hallucination in business contexts: what happens when an AI gives a confident wrong answer in a customer-facing product
- Data privacy and security basics: what data goes into an AI tool, where it is stored, what your company's data policies should address
- Regulatory landscape: EU AI Act basics, U.S. executive orders, what compliance looks like for a business deploying AI (high-level)
- The human-in-the-loop principle: when to require human review before AI output becomes action
- Principles for responsible deployment: NIST AI Risk Management Framework (conceptual overview)

Activity: "Red Team the Product" — teams swap their AI product pitches from Session 1 and spend 20 minutes identifying the top 3 ethical, security, or reliability risks in the other team's product. They present their findings; original teams respond.

Readings (before Session 1):
- Assigned briefing: NIST AI Risk Management Framework — Executive Summary (provided via LMS)
- Assigned article: "When AI Gets It Wrong: Case Studies in AI Failure" — instructor-curated (provided via LMS)

Assignment Due: **Reflection Post 4** — 200–300 words: What is the biggest responsible AI risk in your team's capstone project, and what design choice would you make to mitigate it?

Week 7: Strategy, Synthesis, and What Comes Next

Theme: Pulling it together. How do leaders make decisions about AI adoption? What does it mean to work, lead, and grow in an AI-augmented world?

Session 1 — AI Strategy for Organizations

Topics:
- How organizations think about AI: strategy vs. experimentation vs. transformation
- The AI adoption spectrum: augmentation, automation, and autonomy — and how organizations move along it
- Building an AI roadmap: how to identify priorities, evaluate tools, build for change management, and measure impact
- Common failure modes of enterprise AI: wrong problem, wrong tool, no change management, ignoring ethics — case examples
- The human side: what jobs change, which skills grow in value, how to position yourself as an AI-fluent professional

Activity: Final exam review session and capstone preparation time. Instructor-led Q&A on final exam format; teams finalize their presentations.

Session 2 — Capstone Presentations and Course Synthesis

Topics (brief, 15 minutes):
- Where generative and agentic AI are heading: multimodal agents, physical AI, AI-to-AI economies — brief horizon scan
- The mindset that matters most: curiosity, critical evaluation, and willingness to learn continuously

Activity: **Capstone Presentations** — each team presents their AI workflow or agent (8 minutes) followed by 4 minutes of Q&A from peers and instructor. Presentations are evaluated on: problem clarity, solution design, AI workflow logic, responsible AI considerations, and pitch quality.

Assignment Due:
- **Final Conceptual Exam (take-home, open-note, individual)** — released after Session 1, due 48 hours later. Format: 3 case-based questions on AI strategy, governance, and ethics. No coding. Approximately 90 minutes.
- **Capstone Team Deliverable** — slide deck submitted to LMS before Session 2.

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Assessments, Capstone, and Course Policies

Assessment Overview

This course uses a hybrid assessment model designed to evaluate both **conceptual understanding** and **applied thinking**. No assessment requires coding or a working software prototype. Exams are case-based and test your ability to reason about AI systems, evaluate business decisions, and apply frameworks from the course. The capstone is a team project culminating in a live pitch at the **AI Innovation Showcase** — a trade show–style event open to external practitioners and guests.

Assessment Component	Weight	Format	Due
Reflection Posts (4 total)	20%	Individual, 200–300 words each	Bi-weekly (Weeks 2, 3, 5, 6)
Midterm Conceptual Exam	25%	Individual, take-home, open-note	Released Week 4 Session 2; due Week 5 Session 1
Final Conceptual Exam	25%	Individual, take-home, open-note	Released Week 7 Session 1; due 48 hours later
Team Capstone — AI Innovation Showcase	25%	Team of 3–4, solution pitch (no prototype required)	Week 7 Session 2
Participation and In-Class Build Sprints	10%	Individual	Ongoing

Total: 100%

Reflection Posts (20%)

Four short reflection posts, submitted individually via the course LMS. Each post is 200–300 words and responds to a specific prompt tied to that week's activities (prompts are listed in the weekly schedule). Reflections are not summaries — they are your honest, specific reaction to something that surprised, confused, or interested you. Grading is based on specificity and engagement, not on having the "right" answer.

Grading criteria (each post, 5 points):
- 5 pts: Specific, insightful, connects to course concepts or personal/professional context
- 4 pts: Specific but surface-level connection to concepts
- 3 pts: General observations without specific examples
- 2 pts: Submitted but minimal engagement
- 0 pts: Not submitted

Late posts are accepted up to one week late at half credit. Posts submitted more than one week late receive no credit.

Midterm Conceptual Exam (25%)

Format: Take-home, open-note, individual. Released after Week 4 Session 2; due before Week 5 Session 1.
Estimated time: 60–90 minutes
No coding required. This exam does not ask you to write code, build workflows, or use any tools during the exam itself.

The midterm consists of **four case-based short-answer questions**, each accompanied by a brief scenario (1–2 paragraphs describing a business situation involving AI). You will be asked to analyze the situation, apply a framework or concept from the course, and recommend a course of action with reasoning.

Topics covered:
- How large language models work and why they behave the way they do (hallucination, context limits, temperature)
- Distinguishing generative AI from agentic AI and other AI paradigms
- Identifying automation opportunities using the frequency × effort × standardizability framework
- Evaluating AI tools for a specific business function
- Recognizing responsible AI concerns in a described deployment

Grading: Each question is scored on a 25-point rubric evaluating: accuracy of concept application (10 pts), quality of reasoning (10 pts), and clarity of communication (5 pts).

Academic integrity note: You may use your notes, readings, and course materials. You may use AI tools to check your understanding or review definitions — but your written responses must be your own analysis. Submitting AI-generated answers as your own constitutes academic dishonesty. See the AI Use Policy section below.

Final Conceptual Exam (25%)

Format: Take-home, open-note, individual. Released after Week 7 Session 1; due 48 hours later.
Estimated time: 60–90 minutes
No coding required.

The final consists of **three case-based questions**, each longer and more integrative than the midterm questions. Scenarios will require you to draw on multiple weeks of course content and make a structured recommendation as if you were advising a business leader.

Topics covered:
- AI strategy: how organizations prioritize and sequence AI adoption
- Agentic AI systems in business: appropriate scope, guardrails, and human-in-the-loop design
- Responsible AI: identifying bias, hallucination risk, security exposure, and governance gaps in a described system
- Vendor and platform evaluation: what questions to ask, what tradeoffs to weigh
- The future of work: how AI changes skill requirements and organizational design

Grading: Each question scored on a 33-point rubric: concept integration (13 pts), reasoning quality (13 pts), communication clarity (7 pts).

Team Capstone: AI Innovation Showcase (25%)

Overview

The capstone is the creative center of this course. Working in teams of 3–4, you will identify a real business problem, design an AI-assisted solution, and pitch it live at the **AI Innovation Showcase** — a trade show–style event held during Week 7, Session 2.

No working prototype or live demo is required. Your team's job is to think like an AI product designer or business strategist: define the problem clearly, design a solution thoughtfully, explain how the AI would work in practice, and articulate the business value and risks. The Showcase evaluates the quality of your thinking, not your ability to build software.

The Showcase is open to external guests — business practitioners, entrepreneurs, and industry professionals invited by the instructor. In trade show fashion, guests rotate through team stations during the session, spend 8–10 minutes with each team, ask questions, and move on. Every team presents multiple times to different visitors. This format is deliberately different from a panel presentation: you will give your core pitch several times, adapting it to different audiences, and fielding unpredictable questions from people with real professional experience. That is the skill.

Team Formation

Teams of 3–4 students form during Week 4, Session 2. Students may self-select teams; the instructor will assign remaining students to teams by the end of Week 4. Teams should aim for diversity of business concentration — a team with students from marketing, finance, and operations will typically produce more interesting and realistic work than a team drawn from a single major.

Capstone Milestones

Milestone	Due	Deliverable
Team Registration	End of Week 4	Team names + members submitted via LMS
Problem Statement	Week 5, Session 1	1 paragraph: the business problem and who it affects
Solution Design Sketch	Week 5, Session 2	1-page diagram or description of your proposed AI workflow
Peer Pitch Practice	Week 6, Session 2	3-minute practice pitch to one other team; written peer feedback exchanged
Final Slide Deck	Week 7, before Session 2	5–7 slides submitted to LMS before the Showcase begins
AI Innovation Showcase	Week 7, Session 2	Live pitch and Q&A with rotating external guests

What Your Pitch Should Cover

Your team's presentation at the Showcase is a structured 3-minute pitch, repeated for each visiting guest group. It should be conversational, not read from slides. The five things every strong pitch covers:

1. **The problem** — What business problem are you solving? Who experiences it, and how often? Why does it matter?
2. **The AI solution** — What does the AI do in your solution? Which tool category would power it (conversational AI, no-code automation, agentic workflow)? Walk the guest through the logic of the workflow even without showing it running.

3. **The business value** — What changes for the business or the people using this? Time saved, revenue generated, errors reduced, experience improved?
4. **What could go wrong** — What is the biggest risk or limitation of your solution? What would you do about it? (This is not a weakness — it is evidence of mature thinking.)
5. **The "what's next"** — If this were real, what would the first 90 days of deployment look like?

Guests will ask follow-up questions after your pitch. You do not need to have every answer. "That is a great question — here is how we thought about it, and here is what we are not sure about yet" is a legitimate and impressive response.

Slide Deck Requirements

5–7 slides submitted via LMS before the Showcase. Suggested structure:

- Slide 1: Team name, course, problem headline (one sentence)
- Slide 2: Problem — who, what, how often, why it matters
- Slide 3: Solution overview — the AI workflow described visually (diagram, flowchart, or annotated screenshot of a similar tool)
- Slide 4: Business value — specific, quantified where possible (even estimated)
- Slide 5: Responsible AI — risks and how you would address them
- Slide 6 (optional): What's next — first steps if this were real
- Slide 7 (optional): Team bios or acknowledgments

Slides are a leave-behind for guests, not a script. Design them to be readable at a glance, not dense with text.

Capstone Grading Rubric

Criterion	Points	Description
Problem Clarity	20	Is the business problem well-defined, specific, and meaningful? Does the team articulate who is affected and why it matters?
Solution Design	25	Is the AI workflow logical and well-matched to the problem? Is the tool category choice appropriate? Is the thinking original?
Business Value Articulation	15	Does the team explain concretely what changes for the business or user? Is the value claim plausible and specific?
Responsible AI Awareness	15	Does the team identify at least one meaningful risk and address it credibly?
Pitch Quality and Q&A	25	Is the pitch clear, confident, and adapted to the audience? Does the team handle guest questions thoughtfully?
Total	**100**	

External Guest Judge Packet

External guests at the Showcase receive a one-page briefing that includes: a short description of the course, the rubric above (simplified), and four prompt questions to use with each team:

1. What problem does this solve, and who benefits most?

2. Walk me through what the AI would actually do — step by step.
3. What could go wrong with this in a real business, and how would you handle it?
4. If you had budget and a team, what would you build first?

Guests provide qualitative feedback and engage authentically — they are not scoring teams. The instructor scores the capstone using the rubric above. Guests often provide the most memorable feedback of the semester: plan for it, and enjoy it.

Participation and In-Class Build Sprints (10%)

Active participation is expected in every session. This is a hands-on course — attending without engaging does not meet the participation standard. Participation credit is earned through:

- Contributing to class discussions, activities, and peer feedback moments
- Completing in-class build sprints (Build Sprint 1 in Week 3; Build Sprint 2 in Week 5)
- Showing up prepared: readings done, tools set up, ready to experiment

Build sprints are completed during class time. If you miss a session during which a build sprint occurs, you may complete an equivalent individual task submitted within one week for partial credit (up to 7/10 points for that sprint).

AI Use Policy

This course is about AI. You will use AI tools constantly, intentionally, and as a core part of the learning experience. The AI use policy for this course is therefore more nuanced than a blanket permission or prohibition.

AI is encouraged for:
- Exploring ideas, generating drafts, testing prompts, and experimenting with tools during class activities and build sprints
- Using course tools (Gemini, NotebookLM, Zapier, n8n, Voiceflow) to complete assignments where specified
- Reviewing concepts, checking your understanding, or generating practice questions when studying for exams
- Brainstorming, ideation, and research at any stage of the capstone project

AI is not a substitute for your own thinking in assessments:
- Reflection posts must represent your own observations and reasoning. Submitting AI-generated posts as your own work is academic dishonesty.
- Exam responses must be your own analysis. You may use AI to review definitions or check recall, but the reasoning, recommendation, and written response must be yours.
- In the capstone, AI tools may help you research and ideate — but the problem selection, solution design, and pitch are your team's intellectual contribution.

Disclosure norm: If you use an AI tool in a way that materially shaped a submitted piece of work (beyond grammar assistance), note it briefly: e.g., "I used Gemini to generate an initial outline, which I substantially revised." This is not penalized — it is professional practice. Undisclosed use that misrepresents the origin of your work is the problem, not AI use itself.

This policy reflects the norm emerging across most industries: AI fluency is expected, and so is integrity about what you contributed versus what AI generated.

Academic Integrity

All students are expected to follow the university's academic integrity policy. Beyond the AI-specific guidance above: do not represent others' work as your own, do not share exam content with students who have not yet completed it, and cite sources in any written work drawing on external material.

Attendance

Two or more unexcused absences will result in a one-letter-grade reduction in your final course grade. Three or more unexcused absences may result in course failure. Because this course depends heavily on in-class activity, attendance is directly tied to learning outcomes — not just a formality. Contact the instructor in advance if you anticipate a conflict.

Accessibility and Accommodations

Students requiring academic accommodations should contact the university's disability services office and provide documentation to the instructor within the first two weeks of the course. The instructor is committed to making all course activities — including build sprints and the AI Innovation Showcase — accessible to all students.

Late Work

Assignments submitted late without prior instructor approval are penalized 10% per day, up to a maximum of 50%. Work submitted more than five days late without approval receives no credit. Exceptions are made for documented emergencies; contact the instructor as early as possible.

End of Syllabus — IDSC 4101: Generative and Agentic AI for Business